

Dima Bshara | ID 213932338 | HW2 - BPE Tokenizer Report

1. Implementation and per-tokenizer training. All three tokenizers share one BPE algorithm: character-level start, iterative frequency-based pair merging, and 32 direct two-word bigram tokens. Tokenizer 1 (domain 1) uses vocab_size=4000; tokenizer 2 and tokenizer 3 use vocab_size=5000. Tokenizer 1's vocab_size was tuned via a 9-value sweep (1000-8000) on domain_1_dev.txt NER F1: F1 peaked at 0.4931 at vocab_size=4000 (vs. 0.4572 at 5000), with no loss in speed or tokens/char (an observed dev-set effect, not guaranteed for the hidden test file). The only behavioral change kept in the final code is a training-speed optimization: instead of rescanning the whole corpus to recount pair frequencies after every merge, the algorithm keeps one running pair-count table plus an index from each pair to the sequences that contain it, updating only the sequences a merge actually changes. This produces identical merges, vocabulary, and encoded output to the naive algorithm (verified by direct comparison) while cutting training time substantially. An earlier attempt to bias merge frequencies toward capitalized/entity-like words was tested and reverted after lowering domain-1 F1 (0.4487 to 0.4300); the final code uses plain frequency counts only. Tokenizer 3 uses the identical algorithm, trained via generate_tokenizers.py with --train_files_3 domain_1_train.txt domain_2_train.txt, since it must serve a domain unseen at training time. vocab_size=4000 was also tried for tokenizer 3 and rejected: it raised domain-2 F1 only marginally (0.9611 to 0.9646) while lowering domain-1 F1 (0.4695 to 0.4631) and tokens/char efficiency, so it keeps vocab_size=5000 -- the sweet spot did not transfer to the combined corpus.

2. Top-5 / bottom-5 most frequent bigram tokens (of the 32 direct two-word tokens selected per tokenizer):

Tokenizer 1 top: MTV Movie (449), SO MUCH (205), I'm SO (116), GOOD MORNING (112), IM SO (97). Bottom: REALLY REALLY (44), WHY DO (42), MOTHER'S DAY (42), MOVIE AWARDS (37), CAN DO (35).

Tokenizer 2 top: Donald Trump (3526), New York (2181), Hillary Clinton (1215), White House (1180), Donald Trump's (878). Bottom: Keystone XL (65), GOP Congressman (57), GOP Debate (55), NEW YORK (51), RELATED LINKS (28).

Tokenizer 3 top: MTV Movie (493), HUFFPOST HILL (245), SO MUCH (205), I'm SO (116), GOOD MORNING (112). Bottom: IM NOT (49), SO SAD (47), SO EXCITED (45), REALLY REALLY (44), MOTHER'S DAY (43).

3. Dev-set F1 and tokenizer efficiency (measured on the course GPU VM; tokens/char and reconstruction confirmed identical to a separate CPU run):

Tokenizer 1 (domain 1 on domain 1): vocab 4000, speed 306,650 tok/s, tokens/char 0.3221, reconstruction 100%, NER F1 0.4841.

Tokenizer 2 (domain 2 on domain 2): vocab 5000, speed 290,941 tok/s, tokens/char 0.2976, reconstruction 100%, NER F1 0.9624.

Tokenizer 3 (combined on domain 1): vocab 5000, speed 210,747 tok/s, tokens/char 0.3160, reconstruction 100%, NER F1 0.4695.

Tokenizer 3 (combined on domain 2): vocab 5000, speed 330,894 tok/s, tokens/char 0.3178, reconstruction 100%, NER F1 0.9611.

4. Assessment of tokenizer 3 on the hidden domain. No labeled data exists for the hidden domain, so likely performance is assessed with visible-domain generalization probes rather than a direct measurement. Vocabulary overlap: tokenizer 3 shares 74.4% of its vocabulary with tokenizer 1 and 61.4% with tokenizer 2, far higher than the 31.9% the two domain-specific tokenizers share with each other; tokenizer 3 is a genuine blend, not just one domain copied. Cross-domain compression on random 8,000-line dev samples: tokenizer 3 reaches 1.75 tokens/word and 0.05% UNK on domain 1, and 1.87 tokens/word and 0.04% UNK on domain 2, closer to each domain's own specialist than the other specialist is when forced out-of-domain. Combined-tokenizer validation on the two visible NER benchmarks gives F1=0.4695 on domain 1 and F1=0.9611 on domain 2, in the same range as tokenizer 1 (0.4841 at its tuned vocab_size=4000) and tokenizer 2 (0.9624) on their own domains, supporting reasonable performance on a genuinely unseen domain. Caveat: this validates across the two visible domains, not the actual hidden domain, which may differ stylistically from both. One observed weakness: tokenizer 3's fixed 32-slot bigram budget skews toward domain-1-style bigrams, since domain 1 contributes about four times more training lines than domain 2.